

PENTELEU, Romania

WMO: 152840

Lat: 45.6028N

Lon: 26.4097E

Elev: 1633

StdP: 83.18

Time Zone: 2.00 (EUE)

Period: 88-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.4	-15.7	-25.0	0.5	-7.5	-21.1	0.7	-8.7	27.7	-12.0	23.7	-9.1	9.1	320	0.840

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	20.6	14.7	19.2	13.9	17.9	13.2	15.9	19.3	15.0	18.1	14.1	17.1	4.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.6	12.7	17.5	13.7	12.0	16.5	12.9	11.3	15.4	51.0	19.3	47.9	18.1	45.3	17.1	20.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	15.5	13.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-6.8	-4.9	-3.2	1.5	7.3	11.4	12.6	13.2	8.2	4.2	-0.1	-4.7		
	DBStd	8.38	5.14	4.99	4.72	5.61	3.81	3.66	3.24	3.65	4.76	5.22	5.17	5.07		
	HDD10.0	2799	520	418	408	259	99	27	12	15	89	192	303	457		
	HDD18.3	5498	778	652	666	505	342	208	179	160	303	438	552	715		
	CDD10.0	348	0	0	0	4	16	70	93	115	37	13	1	0		
	CDD18.3	4	0	0	0	0	0	1	1	2	1	0	0	0		
	CDH23.3	8	0	0	0	0	0	1	6	1	1	0	0	0		
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0		
(14) Wind	WSAvg	6.2	7.5	7.1	6.7	6.2	5.3	5.3	5.2	4.4	5.7	6.0	7.0	7.7		
(15) Precipitation	PrecAvg	701	33	29	37	55	85	104	102	84	57	48	35	36		
	PrecMax	959	71	73	71	111	178	187	225	176	123	132	82	72		
	PrecMin	419	9	3	5	22	39	44	37	24	13	2	1	7		
	PrecStd	136	17	15	16	27	40	40	49	38	27	30	18	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	8.1	10.0	15.5	18.0	22.2	23.0	22.2	21.2	17.3	12.7	9.3		
		MCWB	0.8	3.9	4.3	8.2	11.3	15.5	15.7	15.6	14.1	8.8	6.0	2.8		
	2%	DB	2.7	5.1	7.5	12.9	15.9	19.4	20.1	20.5	17.8	15.0	10.3	5.8		
		MCWB	0.5	0.9	3.2	7.5	10.3	14.1	15.0	14.6	12.4	8.6	5.4	2.0		
	5%	DB	1.2	3.1	5.3	10.8	14.1	17.7	18.7	19.1	15.9	13.1	8.5	3.6		
		MCWB	-0.4	1.0	2.2	6.6	9.8	13.7	14.1	13.6	11.4	7.5	4.5	1.6		
	10%	DB	-0.2	1.4	3.2	8.9	12.4	16.3	17.2	17.8	14.1	11.2	6.9	1.8		
		MCWB	-1.4	-0.5	1.0	5.0	8.7	12.7	13.5	13.1	10.4	6.8	3.8	0.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	5.4	5.0	9.6	12.6	16.7	16.7	16.9	15.2	12.4	7.9	4.4		
		MCDB	3.1	7.8	8.9	14.6	15.4	20.4	20.5	21.0	19.9	15.4	10.5	6.1		
	2%	WB	1.3	2.6	3.8	8.3	11.4	15.4	15.7	15.7	13.3	10.1	6.8	3.0		
		MCDB	2.5	4.5	6.9	12.0	14.1	18.3	19.1	19.3	16.9	12.9	8.9	4.3		
	5%	WB	0.0	1.4	2.8	7.3	10.3	14.3	14.9	14.6	12.0	8.7	5.8	1.8		
		MCDB	0.7	3.0	5.4	10.5	13.1	17.0	17.6	18.0	15.3	12.2	7.8	3.6		
	10%	WB	-1.2	0.1	1.5	6.0	9.4	13.4	14.0	13.8	11.2	7.6	4.5	0.4		
		MCDB	-0.4	1.0	3.4	8.8	12.0	16.0	16.7	16.9	14.1	10.4	6.2	1.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.4	5.8	6.1	6.5	6.2	6.1	6.0	5.4	5.3	4.9	4.9		
		MCDBR	5.7	6.5	8.1	8.3	8.4	7.2	7.6	6.8	7.1	7.3	6.3	6.6		
	5% WB	MCWBR	4.3	4.5	5.1	5.4	4.7	4.5	4.7	4.3	4.6	5.8	4.2	3.9		
		MCWBR	5.1	5.8	7.3	7.8	7.2	6.9	7.0	6.3	6.3	6.9	5.4	5.2		
(40) Clear-Sky Solar Irradiance	taub	taub	0.258	0.280	0.311	0.356	0.340	0.341	0.352	0.361	0.330	0.303	0.283	0.259		
		taud	2.439	2.391	2.396	2.308	2.397	2.406	2.371	2.342	2.419	2.496	2.515	2.466		
	Ebn at Noon	Ebn at Noon	870	909	923	902	927	924	908	886	889	871	833	830		
		Edh at Noon	68	87	101	121	115	114	117	116	98	78	64	60		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.30	2.12	3.26	4.43	5.57	5.91	6.02	5.45	3.95	2.58	1.42	1.09		
	RadStd	RadStd	0.16	0.25	0.34	0.46	0.47	0.46	0.46	0.39	0.36	0.30	0.14	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)		+0.59	N/A	N/A	+0.66	N/A	N/A	-127	-170	+78	+27

Nomenclature: See separate page